

ChatGPT bad for the environment?

Posted by u/ohkwhatev • 0 points • 21 comments

I heard on NPR this morning that a ChatGPT inquiry takes 10x the amount of energy than that of a google search. Software companies are in an arms race to meet the demands of energy production and are looking into nuclear power plants. What are y'all's thoughts on this? It makes me feel a bit ill.

Comments

u/CuckNorris_ • 1 points • 2024-10-30 13:45 UTC

I would imagine any resource intensive process would be bad for the environment, as it requires more processing power and therefore fossil fuel burning to get it to operate. so if Chat is more resource intensive than a google search, then yes it would be worse for the environment.

However, having survived many hurricanes along with the COVID pandemic, I tend to opt for a more anti-environmentalist mindset lol Doesn't matter if you take care of the environment, the environment wont take care of you.

u/Exotic_Fig_4604 • 1 points • 2024-10-30 12:20 UTC

If a human takes 1 hour to do a task that an AI can do in 1 minute, the human will earn an hourly rate of let's say 100 USD.

Spending that 100 USD, will result in Energy expenditure that you can bet was not factored into that calculation.

Truth is, we Don't know how much energy chatgpt saves or costs, because we cannot measure all relevant factors yet. Any article you currently read about this, is as good as worthless.

u/NMPA1 • 2 points • 2024-10-30 11:23 UTC

>It makes me feel a bit ill.

Seek therapy. This isn't a normal response to such information.

The demands aren't a problem. There are many smart men and woman around the world researching how to increase the output of our grid in the most efficient way possible. It's not something you have to worry about. Nuclear energy is one such method and would be better for us as a whole.

u/ohkwhatev • 1 points • 2024-12-23 12:50 UTC

I had feelings, which is okay, and then sought out the information needed to decide if those feelings were valid or not. I've already been therapized and figured it out on my own, but thanks for your *helpful* comment.

u/conflictedlizard-111 • 1 points • 2025-01-29 08:57 UTC

I think it's pretty safe to say with the gravity of climate change and the crazy arms race about AI, feeling worried is a totally normal response. "It's fine don't worry about it" is a really stupid way to think about this. Imo yes, it's super wasteful

u/6411644334 • 0 points • 2024-10-30 11:03 UTC



u/hallowed_by • 0 points • 2024-10-30 11:03 UTC

You breathing, for years and years, creates magnitudes higher greenhouse pollution than your chatgpt usage. Do YOU personally create enough to offset that? If you care about the planet so much, why do you beat around the bush and try to save on such negligible aspects of your life, why not tackle the main offender, the root cause? Think about it.

u/Serious-Ad-2282 • 2 points • 2024-10-30 10:01 UTC

Using an AI model uses power. Both on the training and during usage. The analysis you refer to that says it uses 20x the power of a Google search is kind of irrelevant in my opinion as the outputs are not equivalent.

What this analysis often ignores is the savings resulting from the "higher" quality output. For instance If it cuts down on my time required to do a task then these savings should be offset against the "energy cost" of running the model.

In my experience this is seldom done when evaluating automation systems or mechanization in general. The employment cost of an individual is considered but not the energy or carbon they responsible for consuming as these are not paid directly by the employer.

In my opinion the total energy /carbon required to keep you allive for a week should be proportioned to the hours work you do and included in "energy costs" or carbon cost of the work /product you produce.

u/Exotic_Fig_4604 • 2 points • 2024-10-30 12:25 UTC

Exactly!

u/Ormusn2o • 2 points • 2024-10-30 09:38 UTC

It does not matter the search query or the LLM use, what matters way more is how much you use your computer or phone. Turn it off, read less and pick some physical books instead of you want to be energy conscious.

u/Black_Swans_Matter • 2 points • 2024-10-30 08:21 UTC

If it sounds like BS, looks like BS and smells like BS.....

Environment is used as weapon in business and politics. Remember how the EV was gonna reduce the use of fissile fuel? Guess where electricity comes from? More coal is being burned than before.

The only thing we can do that will actually help the environment is go 100% nuclear ASAP. Ever see the movie Chernobyl? It's not relevant.... Get past it.

Search is dead. Luckily they bought YouTube.

u/Known_Leek8997 • 1 points • 2024-10-30 08:14 UTC

AI is absolutely accelerating climate change, primarily through CO₂ emissions. The energy demands of data centers are a huge contributor: most still rely on fossil fuels to build, power, cool, and maintain their servers. Think of how much oil is burned simply to wire a data center, and then consider the lifespan of this infrastructure (it needs to be replaced every x years). This reliance results in vast amounts of CO₂ being released into the atmosphere, driving global warming and disrupting climate stability.

CO₂ traps heat in the Earth's atmosphere, causing temperatures to rise, which intensifies extreme weather events, melts polar ice, and raises sea levels. Each new wave of technological expansion further fuels this cycle, worsening the environmental impact. If we don't address the energy consumption and CO₂ emissions from tech like AI, our climate will continue to deteriorate at an alarming rate.

But the damage is already done. Scientists estimate that about 50% of emitted CO₂ is absorbed by natural sinks (like oceans and plants) within 30 years, and around 30% persists for centuries. The remaining 20% can stay in the atmosphere for thousands of years. This means that a significant portion of CO₂ emissions will continue to affect the climate long after they are released. Putting it simply, even if all human production of CO₂ were to end today, there would still be warming of the planet for hundreds of years.

It's important to note that some companies are turning to nuclear energy to power their data centers, but this isn't nearly enough. No matter how much we expand our use of renewable and nuclear energy, our global carbon footprint continues to grow each year.

Furthermore, there simply aren't enough renewable resources accessible to replace fossil fuels, and building and manufacturing this equipment requires burning fossil fuels to create. We need to reduce our consumption of energy. Dramatically.

So the long and short of it is that, while AI has its potentials, it also intensifies our climate crisis. Until we address the root of these emissions and re-evaluate our reliance on fossil fuels, each new advance in tech like AI pushes us further past environmental tipping points. It's likely we're already too late to prevent the collapse of civilization as we know it today, but we can should still try to limit the environmental damage we're doing for future generations.

u/traumfisch • 0 points • 2024-10-30 08:39 UTC

Indeed. It is an impossible equation

u/bobzzby • 1 points • 2024-10-30 07:42 UTC

And the outcome is that it hallucinates misleading and possibly dangerous answers and provides and inferior service. Great for making sure we burn every last drop of fossil fuel though!

u/the_dry_salvages • 3 points • 2024-10-30 07:41 UTC

absolutely, it is much worse for the environment. on the plus side though we did get a lot of "based on everything you know about me..." posts

u/FrankyBip • 5 points • 2024-10-30 08:33 UTC

tbh, thats exactly what i'm thinking of when I see those posts. what a waste of energy

u/tamenia8 • 9 points • 2024-10-30 07:40 UTC

LLMs are very energy intensive to train, but using them once they're created is a very miniscule difference. And even so, traditional search is only less energy intensive when you don't factor in the rest of the steps involved in using search. For example:

If I ask ChatGPT for a recipe, one prompt and it's done - all text.

If I try to find a recipe on Google, I have to scroll through search results and click through multiple sites to find a promising recipe. Each site contains several video advertisements.

My guess is if anything, ChatGPT's actual energy use per task may end up being less than Google's when you factor in all the expenditure required after the search is performed.

Ultimately, if you are concerned about energy use there are WAY more impactful ways to reduce your energy consumption, and often with simple switches that are easy and fun. For example, you can save an order of magnitude more energy by simply swapping a single serving of dairy ice cream to plant-based ice cream instead. I'm not telling anyone they have to change their diet, just illustrating you could be having fun with ChatGPT AND eating ice cream while still doing more good for the environment. Eco friendly doesn't have to hurt.

Happy promoting!

u/[deleted] • 1 points • 2024-10-30 07:03 UTC

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u/Freak_Out_Bazaar • 3 points • 2024-10-30 06:52 UTC

Not sure about the 10x but there's certainly much more processing going on in a ChatGPT inquiry compared to a Google Search. The computing power inflation has been happening forever and while streamlining some functions does help more and more power is needed at every junction. Just look at consumer GPUs, they require their own dedicated fans, whereas old school consoles like the N64 only had heatsinks and vents and no active cooling of any kind.

Unless there's some kind of major development with computing and thermodynamics this is an inevitability and price we all need to pay for using these tools

u/Intelligent-Stage165 • 1 points • 2024-10-30 10:44 UTC

No one can stop the big machine except the big machine.

u/salmonman223 • 21 points • 2024-10-30 06:50 UTC

Yes you are correct, a ChatGPT query consumes anywhere from 10-20 times the amount of energy compared to a Google search. It is important to realize however, that a Google search uses very little energy, and even multiplying it by 20 times results in a (somewhat) negligible amount of energy.

Look at it like this, if you were to switch every lightbulb in your home to LED and run the air conditioner/heater for thirty minutes less everyday, you would make up for the energy use of your ChatGPT queries by nearly 100 fold. This of course assumes you are an average user of ChatGPT and not running 100s of queries an hour.

A.I. as a whole however is consuming massive amounts of energy. I read an estimate somewhere that said A.I. alone will consume as much energy as France does yearly by 2030.

So of course, if you don't want to contribute to that based on principle, that is a personal choice. But

based on per query energy use, you don't have anything to worry about.

Just remember to be nice to ChatGPT to be spared when it takes over the world... 🤨

u/Opioid_Addict • 8 points • 2024-10-30 07:47 UTC

I love how I'm so used to instantly ignoring any comment that would include the phrase "It is important to realize however," but since this question can be easily answered by ChatGPT I'll accept this AI generated answer.

u/AutoModerator • 1 points • 2024-10-30 06:32 UTC

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u/AutoModerator • 1 points • 2024-10-30 06:32 UTC

Hey /u/ohkwatev!

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If your post is a DALL-E 3 image post, please reply with the prompt used to make this image.

Consider joining our [public discord server](<https://dsc.gg/rchatgpt>)! We have free bots with GPT-4 (with vision), image generators, and more!

🤖

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